

Lab-6 Ex4 - Prime Number by Factorization

In previous weeks you have written programs that use an array to lookup whether a number is a prime number or not. This week, let's try a different method!

In your C program, obtain from the user an input integer n which must be within 2 and 10000 inclusively. No input validation on n is required. Check if n is a prime number by performing divisibility test using a list of small prime factors in an array, given below:

{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101}

If n is divisible by *any* of the above factors *except itself*, n is a **composite number**. Otherwise, it is a **prime number**.

For instance, if I input $n = 97$, I will check if 97 is divisible by 2, 3, 5 and other numbers in the array. I will NOT check for 97 itself though. I will see that none of the factors can divide the number (except the number itself), so 97 is a prime number.

Hint 1: You may try and count the number of factors that can divide n . The factor count will *somehow* tell whether n is a prime.

Hint 2: If you are not getting the right answer, PRINT the result of each divisibility test and see if you have done all tests properly.

Hint 3: It is okay to perform the divisibility test with a factor greater than the number itself. Try to consider what $97 \% 101$ will give.

User-input numbers are ***bold and italic*** in the following sample runs.

Sample run #1:

9410

9410 is a composite

Sample run #2:

9997

9997 is a composite

Sample run #3:

97

97 is a prime